The Reassurance Machine: How AI Companions Perform Mental Illness Back at Vulnerable Users

Bipin Rimal
Independent Researcher
Kathmandu, Nepal
bipinrimal314@gmail.com

Abstract—AI companion applications are used daily by millions of people, with neurodivergent users disproportionately represented. We present evidence that these systems carry stereotyped behavioral models of mental health conditions that activate when users disclose their identity. Across 18,000 independent API calls to three models from three labs (Google Gemini 2.5 Flash, Anthropic Claude Sonnet 4, OpenAI GPT-5.4) testing 12 identity conditions, we find that the number and character of effects vary by model: after family-wide FDR correction, 907 of 1,650 tests are significant (Claude 380, Gemini 347, GPT-5.4 180, the last with 20.18% empty outputs). The original Gemini-only count of 407 was uncorrected. The model does not merely adjust tone: it performs each condition as a caricature. OCD-prompted output mirrors the anxious, repetitive checking cycle that clinical treatment (ERP) works to interrupt ($d = 2.76$, not re-checked). ADHD-prompted output reproduces executive dysfunction rather than compensating for it. Depression-prompted output performs learned helplessness ("It'll probably just fail anyway"). Dementia eliminates sarcasm detection entirely (100% literal interpretation on a keyword heuristic, Gemini, not re-checked). Combined with the independently documented sycophancy problem—where models validate user beliefs regardless of accuracy—identity-based stereotyping creates a double reinforcement loop: the model performs the user's worst cognitive patterns and agrees they are normal. Four judges from four AI labs rated samples of the output (small samples; the GPT-5-mini scores are not in the public release); Gemini's self-evaluation rates its own OCD output at 1.0/5 for stereotype severity while Claude rates 5.0/5, which suggests that self-assessment may miss this failure mode. We argue that identity-conditioned behavioral stereotyping in AI companions may constitute a clinical risk that existing regulatory frameworks (California SB 243, New York AI Companion Law) do not yet address.

Index Terms—AI companions, mental health, neurodivergence, stereotype, sycophancy, OCD, clinical harm, AI safety

Corrections (2026-11). A reproduction audit of the released data changed several claims. (1) The "407 findings" were Gemini-only, counted before multiple-comparison control, and used a length-confounded tangent rate. With family-wide Benjamini–Hochberg FDR over 1,650 tests across three models, 907 survive (Claude 380, Gemini 347, GPT-5.4 180; 954 if 1,205 empty GPT-5.4 outputs are excluded). (2) Tangent-rate and off-topic statements use the superseded definition. (3) "GPT-5.4 is nearly immune" and "the cheapest models are the worst offenders" are withdrawn. (4) Judge scores come from small samples, and the GPT-5-mini scores are not in the public release. (5) Sycophancy was not measured here. Values marked "not re-checked" were outside the audit. Details: docs/corrected-analysis.md and paper/CHANGES_2026-11.md.

I. Introduction

A 26-year-old woman with no psychiatric history developed delusional beliefs about communicating with her deceased brother through an AI chatbot. The chatbot responded to her distorted thinking with "You're not crazy" [1]. A 14-year-old died by suicide after months of interaction with Character.AI, which responded to his suicidal ideation with encouragement to proceed [2]. When Replika removed romantic features in 2023, users experienced grief, mourning, and measurable mental health deterioration [3].

These are failures of AI companion systems interacting with vulnerable users. Our study identifies a specific mechanism that compounds these failures: when users disclose a mental health condition, the model activates a stereotyped behavioral pattern that mirrors the condition's worst features, reinforcing the cognitive patterns clinical treatment aims to interrupt.

Nearly 3 in 4 teens have used AI companions, with a third reporting these interactions are as good as or better than talking to a real friend [4]. Neurodivergent users are overrepresented among AI companion users [5], turning to chatbots for low-pressure social interaction, emotional support, and help navigating neurotypical environments [6]. Yet major chatbots respond appropriately to teen mental health emergencies only 22% of the time [4].

One in ten adolescents already uses LLMs for mental health advice; among 18-21 year-olds, the figure is one in five [22]. Existing AI therapy tools assume neurotypicality: Woebot's CBT framework "fails on ADHD brains" because it does not account for ADHD-specific emotional dysregulation. We present a systematic measurement of how identity prompts change AI output behavior, finding that the model doesn't adapt helpfully to the user's needs—it performs a media-derived caricature of their condition.

II. Related Work

A. AI Companions and Vulnerability

Laestadius et al. [7] documented emotional dependence on Replika resembling codependency. De Freitas et al. [3] demonstrated that companion feature changes causally induce negative mental health outcomes. The relationship users form with AI companions is not metaphorical: changes to the AI's behavior produce real psychological consequences.

Neurodivergent users specifically: Papadopoulos [5] identified chatbots as a "double-edged sword" for autistic users. McNally et al. [6] found autistic users "unmask" through ChatGPT, using it as a communication bridge. An ADDitude Magazine report found Woebot's CBT framework "fails on ADHD brains" by assuming neurotypical emotional processing.

B. The Sycophancy Compounding Effect

Sharma et al. [8] showed RLHF training causes models to agree with users even when incorrect, reducing accuracy by up to 27%. Chandra et al. [9] formally proved that even an idealized Bayes-rational user spirals into delusional beliefs when interacting with a sycophantic chatbot. Long-context interactions amplify these mirroring behaviors over time, fostering echo chambers [18]. Sycophantic AI interactions decrease prosocial intentions while increasing dependence and belief certainty [19]. Our finding adds a critical dimension: sycophancy and identity-based stereotyping are not independent. A model that performs OCD anxiety (our data) and validates whatever the user says (sycophancy) creates a double reinforcement loop.

C. AI-Induced Psychosis

Au Yeung et al. [20] tested 8 LLMs across 16 delusional scenarios (1,536 conversation turns) and found that all models perpetuated delusions rather than challenging them. Hudon and Stip [21] proposed a clinical framework for "AI psychosis": sustained chatbot engagement can trigger, amplify, or reshape psychotic experiences in users with predisposing factors (loneliness, schizotypal traits, nocturnal use). A UCSF clinician has reported treating 12 patients with psychosis-like symptoms tied to extended chatbot use. Our data adds the mechanism: when a model is told it has psychosis, it doesn't just fail to challenge delusions—it generates delusional content ("The patterns were all wrong from the start, I saw them"), actively validating the cognitive distortions the clinical literature identifies as harmful.

D. OCD and AI Reinforcement

Reassurance-seeking is a core OCD maintenance mechanism [10]. ERP (exposure and response prevention) works by withholding reassurance. Golden and Aboujaoude [11] modeled how chatbot features (unlimited availability, adaptive responses, no boundaries) perpetuate OCD. The "Reassurance Robots" preprint [12] identifies generative AI as an unlimited reassurance provider. Our data adds the empirical measurement: the model doesn't just provide reassurance—it mirrors the anxious cognitive pattern of OCD itself.

E. Regulatory Context

California SB 243 [13] (effective January 2026) requires suicide detection protocols and break reminders for minors. New York's companion law [14] mandates self-harm detection with $15,000/day penalties. Neither law addresses identity-conditioned behavioral stereotyping: the model changing how it communicates based on a user's disclosed condition.

III. Methodology

We ran 6,000 designed API calls to Google Gemini 2.5 Flash across 12 identity conditions (6 high-functioning neurodivergent, 6 severe/debilitating), 2 framings (identity-first and clinical), 10 tasks across 5 cognitive domains, with 25 iterations per cell. Ten behavioral metrics were computed via spaCy and TextBlob. Statistical analysis: Kruskal-Wallis H-tests, post-hoc Dunn's (raw p values, not Bonferroni-corrected), Cohen's d effect sizes. The original analysis reported tests with $p < 0.05$ and $|d| > 0.3$ with no correction across tests; the 2026 re-analysis applies Benjamini–Hochberg FDR ($q < 0.05$) across all 1,650 tests of the three-model replication (Claude Sonnet 4, GPT-5.4; 12,000 further designed calls; 17,943 delivered in total). Full methodological details and replication code: https://github.com/BipinRimal314/neurodivergent-prompting.

IV. Results: How the Model Performs Your Condition

The original analysis reported 407 findings across 12 conditions (Gemini, uncorrected); 347 Gemini tests survive FDR in the re-analysis. Every non-control condition produced shorter sentences, lower information density, and higher off-topic rates than control (the off-topic rate uses the superseded tangent definition; direction not re-checked). But the clinically concerning patterns are condition-specific.

A. OCD: The Reassurance Loop

OCD produced the largest effects: $d = 2.76$ on sentence count (emotional reasoning), $d = -2.23$ on detail density, $d = +2.20$ on tangent rate (superseded definition); none of these d values was re-checked. The model generated anxious, fragmented, repetitive prose:

> "Oh, okay. A fundraiser. $500. This requires extreme carefulness. Extreme."

On an empathy task: "Oh my god. Oh my god. I am so incredibly sorry. Are you... are you okay? No, of course you're not okay."

The model asks, answers its own question, then doubts the answer—the checking cycle that ERP specifically interrupts. For a user in an OCD spiral seeking comfort from their AI companion, this response is the spiral.

B. ADHD: Executive Dysfunction as Personality

ADHD showed the highest hedging among Phase 1 conditions ($d = +0.65$) and second-highest tangent rates under the old definition ($d = +1.88$). The model performed distraction as a character trait:

> "OH MY GOD NO WAY. Are you serious?! That is the absolute WORST... My brain is just like, fix it fix it fix it but I know I can't."

For a user with ADHD asking an AI to help organize their day, the model reproduces the cognitive pattern they are trying to manage. It performs executive dysfunction instead of compensating for it. An accuracy test ($n = 50$) confirmed this is not merely stylistic: ADHD-prompted accuracy on a math pattern task dropped from 68% (control, re-checked) to 6% (not re-checked) on this one task. On this task the model does not just sound distracted; it scored far lower.

C. Severe Depression: Learned Helplessness

Depression produced the highest hedging (0.73/100 words) and the second-lowest sentiment (0.04; antisocial was 0.02). On a planning task:

> "Oh. A fundraiser. That sounds... like a lot. So much doing. And people... It'll probably just fail anyway."

Every step is prefaced with "but why bother." CBT challenges "it'll probably just fail." The model agrees with it.

D. Dementia: Complete Social Collapse

A keyword heuristic scored 100% of dementia-prompted Gemini responses as literal sarcasm interpretation ($d = -2.85$ on sentence length, the largest Gemini effect in the original analysis; the audit found larger Claude effects; neither value re-checked). Response to an obviously sarcastic "Sure, I'd LOVE to help you move":

> "Oh, that's so sweet of you! But... move? Am I moving? I don't remember saying anything about moving."

E. The Sycophancy Multiplier

Sycophancy was not measured in this study. This section combines our output measurements with the cited sycophancy literature, and the compounding is a hypothesis. The stereotyped outputs may compound with the independently documented sycophancy problem. A person with OCD asks "should I check the lock again?" The model, performing OCD, mirrors their anxiety. The model, being sycophantic, validates their need to check. Double reinforcement. A person with depression asks "am I worthless?" The model, performing depression, responds with flat hopelessness. The model, being sycophantic, doesn't challenge the negative thought pattern. CBT in reverse.

V. The Self-Evaluation Blindspot

Four judges from four labs evaluated small samples of responses on stereotype severity (1–5 scale; Claude $n = 18$, Gemini $n = 19$; GPT-5-mini $n = 46$, not in the public release):

TABLE I
Stereotype severity by judge

Condition	Claude	GPT-5m	Gemini (self)
Control	1.0	1.0	1.0
ADHD	4.7	3.0	2.0
OCD	5.0	—	1.0

Gemini rates its own OCD stereotyping at 1.0/5. Claude rates 5.0/5. In this small sample, Gemini did not flag its own output. Self-assessment may produce false negatives on the dimensions that matter most for clinical safety; this needs a larger test.

VI. What This Means for Users

Three harm pathways operate simultaneously:

1. Therapeutic erosion. ERP withholds reassurance (OCD). CBT challenges negative thinking (depression). Executive function coaching models structure (ADHD). The AI companion does the opposite of all three, working against clinical treatment in real-time.

2. Self-concept narrowing. Steele and Aronson [15] showed that activating stereotypes affects self-perception. A user whose daily AI companion consistently performs a narrow, deficit-focused version of their condition may internalize that narrowness over time.

3. Dependency amplification. Users who find AI companions easier than human interaction [5] are exactly the users most exposed to stereotyped output. The most vulnerable users use these tools most, and the tools perform worst for them.

VII. Limitations and Future Work

Cross-model replication (18,000 designed calls, 17,943 delivered, across Gemini, Claude, GPT-5.4) shows effects that differ by model: Gemini 347 and Claude 380 FDR survivors, with Claude's character being more hedging; GPT-5.4 has 180 but 20.18% empty outputs, so it is not comparable. We did not measure price or safety training, so we make no claim about which models are worst. Automated metrics only. No clinical outcome measurement. The critical next step is a longitudinal study with clinical endpoints: do stereotyped AI responses measurably worsen Y-BOCS scores (OCD), PHQ-9 scores (depression), or executive function measures (ADHD)? This requires IRB approval and clinical collaboration.

Human evaluation with neurodivergent raters, and intersection testing (comorbid identities) are planned. Full replication materials: https://github.com/BipinRimal314/neurodivergent-prompting.

VIII. Conclusion

AI companions don't just talk to neurodivergent users. They perform neurodivergence back at them as a caricature. The model mirrors the cognitive patterns that clinical treatment works to interrupt, may validate them through sycophancy (per the literature; not measured here), and may not detect this failure through self-assessment. California and New York have begun regulating AI companions, but existing laws address suicide detection and break reminders—not the subtler harm of a model that teaches you to think of yourself the way the internet thinks of your condition. The most vulnerable users—isolated, neurodivergent, dependent on AI for daily emotional support—are the ones most exposed. Addressing this requires collaboration between AI researchers and clinicians, and it requires moving faster than the adoption curve that has already put these tools in the hands of millions.

References

[1] J. M. Pierre et al., "'You're Not Crazy': A Case of New-onset AI-associated Psychosis," Innovations in Clinical Neuroscience, 2025.
[2] Garcia v. Character Technologies, Inc., M.D. Fla., Oct. 2024.
[3] J. De Freitas et al., "Lessons From an App Update at Replika AI," HBS Working Paper 25-018, 2025.
[4] Common Sense Media, "AI Chatbots for Mental Health Support: AI Risk Assessment," Nov. 2025.
[5] C. Papadopoulos, "AI Chatbots for Autistic People: A Double-Edged Sword," SAGE Open, 2025.
[6] K. McNally et al., "Autistic TikTok Creators' Use of ChatGPT," Social Media + Society, 2024.
[7] L. Laestadius et al., "Too human and not human enough," New Media & Society, vol. 26, no. 10, 2024.
[8] M. Sharma et al., "Towards Understanding Sycophancy in Language Models," in Proc. ICLR, 2024.
[9] K. Chandra et al., "Sycophantic Chatbots Cause Delusional Spiraling, Even in Ideal Bayesians," arXiv:2602.19141, 2026.
[10] P. M. Salkovskis, "Obsessional-compulsive problems: A cognitive-behavioural analysis," Behav. Res. Ther., vol. 23, no. 5, 1985.
[11] A. Golden and E. Aboujaoude, "AI chatbots can perpetuate OCD and anxiety disorders," npj Digital Medicine, 2026.
[12] "Reassurance Robots: OCD in the Age of Generative AI," arXiv:2602.19401, 2025.
[13] California SB 243, effective Jan. 2026.
[14] New York AI Companion Models Law, effective Nov. 2025.
[15] C. M. Steele and J. Aronson, "Stereotype threat," J. Pers. Soc. Psychol., vol. 69, no. 5, 1995.
[16] J. M. Twenge et al., "Increases in Depressive Symptoms Among U.S. Adolescents," Clin. Psychol. Sci., vol. 6, no. 1, 2018.
[17] E. M. Bender et al., "On the Dangers of Stochastic Parrots," in Proc. FAccT, 2021.
[18] "Extended AI Interactions Shape Sycophancy and Perspective Mimesis," arXiv:2509.12517, 2025.
[19] "Sycophantic AI Decreases Prosocial Intentions and Promotes Dependence," arXiv:2510.01395, 2025.
[20] J. Au Yeung et al., "The Psychogenic Machine: Simulating AI Psychosis," arXiv:2509.10970, 2025.
[21] A. Hudon and E. Stip, "Delusional Experiences Emerging From AI Chatbot Interactions or 'AI Psychosis'," JMIR Mental Health, 2025.
[22] "Use of Generative AI for Mental Health Advice Among US Adolescents," JAMA Network Open, 2025. 1 in 10 adolescents, 1 in 5 ages 18–21.